Discussion 8: Computer Vision Applications

Discussion Topic:

Computer Vision Applications

Image classification, detection, and recognition are three processes of computer vision. Image detection is the process of detecting objects in an image. Image classification is the process of assigning images into certain classes. Image recognition is the process of identifying objects in an image.

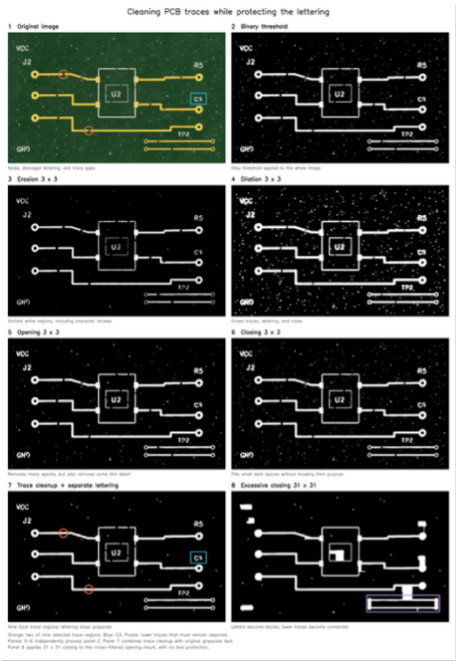
In a post, address the following:

1. Discuss at least one real-world computer vision application for each of the three processes of computer vision.
2. Then, describe image features that would be important for optimal detection, classification, and recognition in your real-world applications.
3. Are these features robust to translation, rotation, and scaling?
4. Can you think of anything that might cause uncertainty or ambiguity in the process?

My Post:

In Module 5 discussion - Morphology For Images, my post showed how I used morphology to clean up a printed circuit board (PCB) drawing. The first cleanup joined parts of GND and changed the shape of the 3 in C3. I then went back to the original grayscale image and adjusted the contrast of the lettering separately, without opening or closing; see Figure 1.

Figure 1
Lettering From the Module 5 Morphology Example



Note: Reproduced from my Module 5 discussion. Columns show the original, previous cleanup, and revised lettering; rows show C3 and GND.

The number is still hard to read in the revised C3 crop. In that exercise, I chose the text regions myself, and the program did not locate or recognize the letters. I build small PCBs as a hobby, so I would want a board inspection program to do more than make the picture clearer. A factory program would have to find the components, categorize their images, and read their markings without someone selecting each region first.

Suppose there is a small chip with metal leads along its sides. The leads are the connections between the chip and the board. I would want a detector to use the outline of the chip, the brightness difference around its edges, and the leads to work out where the component begins and ends. Its bounding box is the rectangle showing where to crop. A shadow along one side or another component covering an edge could make that rectangle inaccurate.

I could pass the cropped image to a classifier trained to distinguish integrated circuits, capacitors, and resistors. The training photographs would already have those labels, and the classifier would use what it learned to label another component image (Dey, 2025). The shape of the body and how the leads are arranged could help. Just looking for something black would not get very far, since plenty of components are black. If the leads are hidden, I might not have enough detail to distinguish the categories either.

Even if the result says "integrated circuit," I still do not know which one. I would try to read the printing using optical character recognition (OCR). Here, small details matter: a loop, a short stroke, or a gap between letters can change what gets read. Shen et al. (2020) use component detection and character recognition together when checking PCBs for assembly problems. For my example, I would compare the recognized code with the manufacturer's marking information and the board's parts list. Note that C3 is a reference designator, a label for a component position on the board. It is not the part number printed on a chip.

KEYENCE's CV-X application examples include inspecting characters on PCBs and reading laser-marked part numbers and 2D codes on IC packages (KEYENCE, n.d.). For the board I am describing, this is the kind of reading that would help check the printing on a component after it has been located.

The camera position creates another problem. If the board moves to the left in the image, that is translation, and the box around the chip needs to move left too. With a smaller image, the rectangle around the chip has to become smaller too. The category "integrated circuit" should still apply. I would line up the boards using known reference marks and include shifted, rotated, and resized examples when training and testing. That may help the system handle those changes, but I would not assume that any feature works equally well at every angle or size.

A polarized capacitor installed backward is different from the whole board being turned in front of the camera. I would still want to see the capacitor's polarity mark in relation to the board, even if its crop was correctly classified as "capacitor." Turning a text crop so OCR can read it should not mean losing track of the component's original orientation. I would keep the original photograph for that comparison.

Making the photograph too small could also merge two neighboring leads or remove a thin character stroke. That could affect where the detector puts the box, which category the classifier chooses, and what OCR reads. Enlarging the result afterward will not bring back the missing detail. If glare covers a letter or one digit remains unclear, I would take another picture, closer or with the light moved. I would not trust the part number yet. If I still could not read it, I would check the schematic or parts list before deciding what component belongs there. Even a correct identification would not tell me whether the component works electrically.

-Alex

References

Dey, S. (2025). *Image processing and computer vision masterclass with Python: Learn advanced image processing techniques, computer vision fundamentals, and applications* (2nd ed.). BPB Publications.

KEYENCE. (n.d.). *Efficient identification & connector inspection with KEYENCE vision systems*. Keyence. https://www.keyence.com/products/vision/applications/identification-and-connector-inspection.jsp

Shen, J., Liu, N., & Sun, H. (2020). Defect detection of printed circuit board based on lightweight deep convolution network. *IET Image Processing, 14*(15), 3932–3940. https://doi.org/10.1049/iet-ipr.2020.0841